Student Activity

The cup filled with ice cubes has tiny water droplets on its outer surface (Figure 6.20). Identify why.

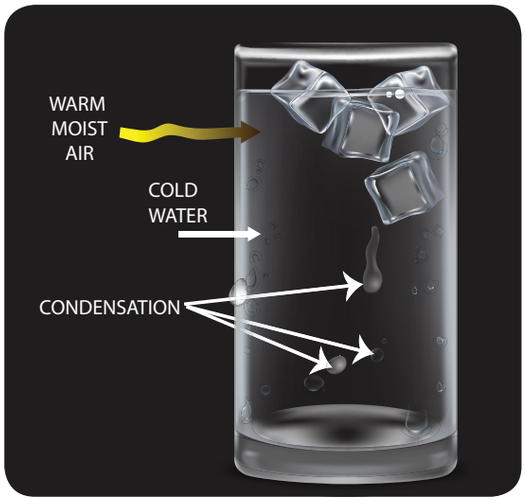


Figure 6.20 Process of Condensation

The moisture in the atmosphere is based on the following processes:

a. Evaporation – Water changes from liquid state to gaseous (vapour) state.
b. Transpiration – Water state changes from liquid in to (gas) vapour state due to the activity of plants.
c. Evapotranspiration – This denotes that the total amount of (liquid) water state changed in to (gas) vapour state due to evaporation and the activity of plants transpiration.

Isonephs – The imaginary line connecting the places having equal amount of cloudiness.

6.5.2 Clouds and its Types

Clouds are tiny water droplets suspended in the air formed due to the condensation (Figure 6.21).

The clouds can be classified based on their form, height and appearance as follows:

a. **High clouds:** Mainly cirrus (Ci) which are feathery form at 6 km above the ground.
   i. Cirrus (Ci) – This looks fibrous and appears as wisps cotton in the blue sky. It indicates fair weather and gives brilliant sun set.
   ii. Cirro Cumulus (Cc) – This appears as white globular masses, forming a mackerel sky.
   iii. Cirro Stratus (Cs) – This resembles a thin white sheet. The sky looks milky and the sun and moon shines through this clouds and form a 'halo'.
b. **Middle Clouds:** Mainly Alto (Alt) clouds at 2 km to 6 km above the ground.
   iv. Altocumulus (Alt-Cu): These are woolly, bumpy clouds arranged in layers appearing like waves in the blue sky. They indicate fine weather.
   v. Altostratus (Alt-St): These are denser and have watery look.
c. **Low Clouds:** Mainly Stratus or sheet clouds below 2 km height.
   vi. Stratocumulus (St-Cu): This is rough and bumpy clouds with wavy structure.
   vii. Stratus (St): This is very low cloud, uniformly grey and thick, appears like highland fog. It brings dull weather and light drizzle. It reduces the visibility and is a hindrance to air transportation.

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